

Training data set: Who buys computer?

age	income	student	credit_rating	buys_computer
<=30	high	no	fair	no
<=30	high	no	excellent	no
31...40	high	no	fair	yes
>40	medium	no	fair	yes
>40	low	yes	fair	yes
>40	low	yes	excellent	no
31...40	low	yes	excellent	yes
<=30	medium	no	fair	no
<=30	low	yes	fair	yes
>40	medium	yes	fair	yes
<=30	medium	yes	excellent	yes
31...40	medium	no	excellent	yes
31...40	high	yes	fair	yes
>40	medium	no	excellent	no

ขั้นตอนที่ 1 Info (Entropy) การหาเอนโทรปี (Decision Tree Inducing) สำหรับ "who buys computer?"

ขั้นตอนที่ 1 หาค่าเอนโทรปีของโหนดราก (Root Node)

ข้อมูลทั้งหมด D จำนวน 14 ตัวอย่าง
 $\frac{9}{14}$ (yes) = 9
 $\frac{5}{14}$ (no) = 5

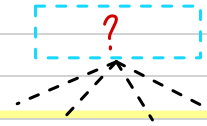
$$\text{สูตร Info}(D) = -\sum_{i=1}^n p_i \log_2(p_i)$$

$$\begin{aligned} \text{Info}(D) &= I(9, 5) \\ &= \frac{9}{14} \log_2\left(\frac{9}{14}\right) - \frac{5}{14} \log_2\left(\frac{5}{14}\right) \\ &= -0.637(-0.637) - 0.357(-1.486) \end{aligned}$$

$$\therefore \text{Info}(D) = 0.940$$

ขั้นตอนที่ 2 การเลือกแอตทริบิวต์เพื่อแยกข้อมูล (Root Split)

ใช้วิธีวัดด้วย Information Gain มาเลือกแอตทริบิวต์ที่จะใช้แยกข้อมูล



แอตทริบิวต์ age มี 3 ค่าคือ

age	buys computer
<=30	no
<=30	no
31..40	yes
>40	yes
>40	yes
>40	no
31..40	yes
<=30	no
<=30	yes
>40	yes
<=30	yes
31..40	yes
31..40	yes
>40	no

	yes	no	D
≤ 30	2	3	5
31 - 40	4	0	4
> 40	3	2	5

$$I(2,3) = -\frac{2}{5} \log_2 \left(\frac{2}{5} \right) - \frac{3}{5} \log_2 \left(\frac{3}{5} \right) = 0.971$$

$$I(4,0) = 0 \quad \text{* "Buy" (yes)}$$

$$I(3,2) = -\frac{3}{5} \log_2 \left(\frac{3}{5} \right) - \frac{2}{5} \log_2 \left(\frac{2}{5} \right) = 0.971$$

$$\text{Info}_{age}(D) = \frac{5}{14}(0.971) + \frac{4}{14}(0) + \frac{5}{14}(0.971) = 0.694$$

$$\therefore \text{Gain}(age) = 0.941 - 0.694 = 0.246$$

แอตทริบิวต์ income มี 3 ค่าคือ

income	buys computer
high	no
high	no
high	yes
medium	yes
low	yes
low	no
low	yes
medium	no
low	yes
medium	yes
medium	yes
medium	yes
high	yes
medium	no

	yes	no	D
high	2	2	4
medium	4	2	6
low	3	1	4

$$I(2,2) = -\frac{2}{4} \log_2 \left(\frac{2}{4} \right) - \frac{2}{4} \log_2 \left(\frac{2}{4} \right) = 1.0$$

$$I(4,2) = -\frac{4}{6} \log_2 \left(\frac{4}{6} \right) - \frac{2}{6} \log_2 \left(\frac{2}{6} \right) = 0.918$$

$$I(3,1) = -\frac{3}{4} \log_2 \left(\frac{3}{4} \right) - \frac{1}{4} \log_2 \left(\frac{1}{4} \right) = 0.811$$

$$\text{Info}_{income}(D) = \frac{4}{14}(1.0) + \frac{6}{14}(0.918) + \frac{4}{14}(0.811) = 0.911$$

$$\therefore \text{Gain}(income) = 0.940 - 0.911 = 0.029$$

แอตทริบิวต์ student มี 2 ค่าคือ

student	buys computer
no	no
no	no
no	yes
no	yes
yes	yes
yes	no
yes	yes
yes	yes
yes	yes
yes	yes
no	yes
yes	yes
no	no

	yes	no	D
yes	6	1	7
no	3	4	7

$$I(6,1) = -\frac{6}{7} \log_2 \left(\frac{6}{7} \right) - \frac{1}{7} \log_2 \left(\frac{1}{7} \right) = 0.592$$

$$I(3,4) = -\frac{3}{7} \log_2 \left(\frac{3}{7} \right) - \frac{4}{7} \log_2 \left(\frac{4}{7} \right) = 0.985$$

$$\text{Info}_{student}(D) = \frac{7}{14}(0.592) + \frac{7}{14}(0.985) = 0.789$$

$$\therefore \text{Gain}(student) = 0.940 - 0.789 = 0.151$$

แอตทริบิวต์ credit_rating มี 2 ค่าคือ

credit_rating	buys computer
fair	no
excellent	no
fair	yes
fair	yes
excellent	no
excellent	yes
fair	no
fair	yes
fair	yes
excellent	yes
excellent	yes
fair	yes
excellent	no

	yes	no	D
fair	6	2	8
excellent	3	3	6

$$I(6,2) = -\frac{6}{8} \log_2 \left(\frac{6}{8} \right) - \frac{2}{8} \log_2 \left(\frac{2}{8} \right) = 0.811$$

$$I(3,3) = -\frac{3}{6} \log_2 \left(\frac{3}{6} \right) - \frac{3}{6} \log_2 \left(\frac{3}{6} \right) = 1.0$$

$$\text{Info}_{credit_rating}(D) = \frac{8}{14}(0.811) + \frac{6}{14}(1.0) = 0.848$$

$$\therefore \text{Gain}(credit_rating) = 0.940 - 0.848 = 0.092$$

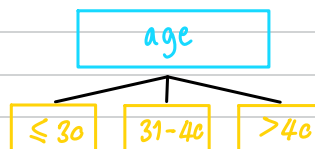
age	income	student	credit_rating	buys computer
<=30	high	no	fair	no
<=30	high	no	excellent	no
31..40	high	no	fair	yes
>40	medium	no	fair	yes
>40	low	yes	fair	yes
>40	low	yes	excellent	no
31..40	low	yes	excellent	yes
<=30	medium	no	fair	no
<=30	low	yes	fair	yes
>40	medium	yes	fair	yes
<=30	medium	yes	excellent	yes
31..40	medium	no	excellent	yes
31..40	high	yes	fair	yes
>40	medium	no	excellent	no

สรุป: Gain(age) = 0.246 (สูงสุด)

$$\text{Gain}(income) = 0.029$$

$$\text{Gain}(student) = 0.151$$

$$\text{Gain}(credit_rating) = 0.048$$

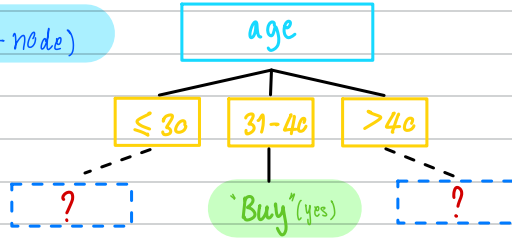


เลือกแอตทริบิวต์ age เป็นแอตทริบิวต์ราก (Root Node) ใช้วิธีวัดด้วย Information Gain สูงสุด (0.246)

Gain: age > student > credit_rating > income

ขั้นตอนที่ 3 คำนวณเอนโทรปี (Sub-node)

หลังจากแบ่งตัว age
ออกมา เอนโทรปีหรือความไม่แน่นอน



	yes	no	ID
≤ 30	2	3	5
31-40	4	0	4
> 40	3	2	5

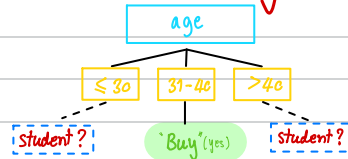
$$I(2,3) = -\frac{2}{5} \log_2 \left(\frac{2}{5} \right) - \frac{3}{5} \log_2 \left(\frac{3}{5} \right) = 0.971$$

$$I(4,0) = 0$$

$$I(3,2) = -\frac{3}{5} \log_2 \left(\frac{3}{5} \right) - \frac{2}{5} \log_2 \left(\frac{2}{5} \right) = 0.971$$

age	income	student	credit_rating	buys_computer
≤ 30	high	no	fair	no
≤ 30	high	no	excellent	no
31...40	high	no	fair	yes
> 40	medium	no	fair	yes
> 40	low	yes	fair	yes
> 40	low	yes	excellent	no
31...40	low	yes	excellent	yes
≤ 30	medium	no	fair	no
≤ 30	low	yes	fair	yes
> 40	medium	yes	fair	yes
≤ 30	medium	yes	excellent	yes
31...40	medium	no	excellent	yes
31...40	high	yes	fair	yes
> 40	medium	no	excellent	no

Gain : ~~age~~ > student > credit_rating > income



1) คำนวณเอนโทรปี student

(1.1) คำนวณเอนโทรปีสำหรับ ≤ 30 info = 0.971

age	student	buys_computer
≤ 30	no	no
31-40	no	yes
> 40	yes	yes
> 40	yes	no
31-40	yes	yes
≤ 30	yes	no
> 40	yes	yes
≤ 30	yes	yes
31-40	no	yes
> 40	no	no

	yes	no	ID
yes	2	0	2
no	0	3	3
			5

$$I(2,0) = 0$$

$$I(0,3) = 0$$

$$Info_{student}(D_{\leq 30}) = \frac{2}{5}(0) + \frac{3}{5}(0) = 0$$

$$\therefore Gain_{student} = 0.971 - 0 = 0.971$$

(1.2) คำนวณเอนโทรปีสำหรับ > 40 info = 0.971

age	student	buys_computer
≤ 30	no	no
≤ 30	no	no
31-40	no	yes
> 40	yes	yes
> 40	yes	no
31-40	yes	yes
≤ 30	yes	no
> 40	yes	yes
≤ 30	yes	yes
31-40	no	yes
> 40	no	no

	yes	no	ID
yes	2	1	3
no	1	1	2
			5

$$I(2,1) = -\frac{2}{3} \log_2 \left(\frac{2}{3} \right) - \frac{1}{3} \log_2 \left(\frac{1}{3} \right) = 0.918$$

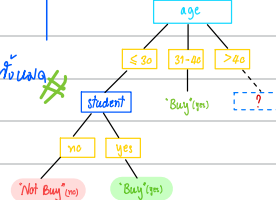
$$I(1,1) = -\frac{1}{2} \log_2 \left(\frac{1}{2} \right) - \frac{1}{2} \log_2 \left(\frac{1}{2} \right) = 1.0$$

$$Info_{student}(D_{> 40}) = \frac{2}{3}(0.918) + \frac{1}{3}(1.0) = 0.9508$$

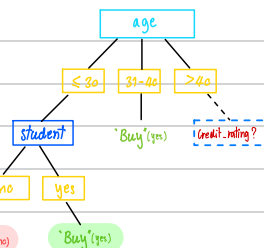
$$\therefore Gain_{D_{> 40}}(student) = 0.971 - 0.9508 = 0.0202$$

วิธีที่ 2 คำนวณ Gain_{student}(student) (0.971) > Gain_{D_{> 40}}(student) (0.020)

สรุป สำหรับ ≤ 30 info = 0.971 สำหรับ > 40 info = 0.9508



Gain : ~~age~~ > student > credit_rating > income



2) คำนวณเอนโทรปี credit_rating

age	credit_rating	buys_computer
≤ 30	fair	no
≤ 30	excellent	no
31-40	fair	yes
> 40	fair	yes
> 40	fair	yes
> 40	excellent	no
31-40	excellent	yes
≤ 30	fair	no
> 40	fair	yes
≤ 30	excellent	yes
31-40	excellent	yes
31-40	fair	yes
> 40	excellent	no

	yes	no	ID
fair	3	0	3
excellent	0	2	2
			5

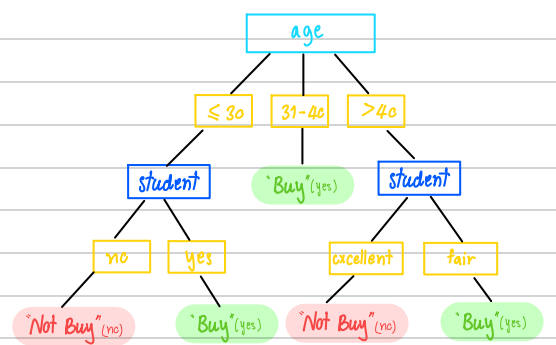
$$I(3,0) = 0$$

$$I(0,2) = 0$$

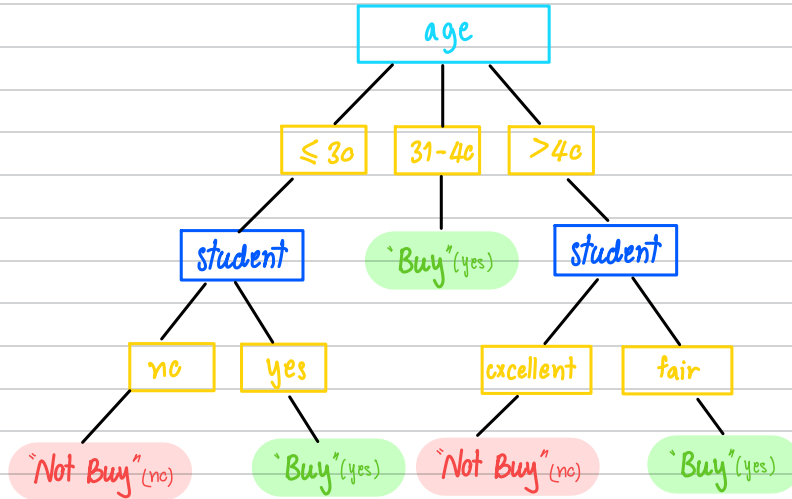
$$Info_{credit_rating}(D_{> 40}) = \frac{3}{5}(0) + \frac{2}{5}(0) = 0$$

$$\therefore Gain(Credit_rating) = 0.971 - 0 = 0.971$$

สรุป สำหรับ > 40 info = 0 สำหรับ ≤ 30 info = 0.971



ขั้นตอนที่ 4 สรุป/นิ



age	income	student	credit_rating	buys computer
≤30	high	no	fair	no
≤30	high	no	excellent	no
31...40	high	no	fair	yes
>40	medium	no	fair	yes
>40	low	yes	fair	yes
>40	low	yes	excellent	no
31...40	low	yes	excellent	yes
≤30	medium	no	fair	no
≤30	low	yes	fair	yes
>40	medium	yes	fair	yes
≤30	medium	yes	excellent	yes
31...40	medium	no	excellent	yes
31...40	high	yes	fair	yes
>40	medium	no	excellent	no

สรุป Info ที่คำนวณ

1. ค่า Entropy Root : $Info = 0.940$
2. ค่า Entropy ของ age : เมื่อ $Info = 0.694$ (ลดเหลือ 0.246)
3. ค่า Entropy ของ age 31-40 : $Info = 0$ (บริสุทธิ์)
4. ค่า Entropy ของ age ≤ 30 : เมื่อ 0.971 หรือ เมื่อแบ่งตาม student แล้ว $Info = 0$ (บริสุทธิ์)
5. ค่า Entropy ของ age > 40 : เมื่อ 0.971 หรือ เมื่อแบ่งตาม credit_rating แล้ว $Info = 0$ (บริสุทธิ์)